Odd kernels, two characters, and cyclotomic polyhedra

human_compression project — private review edition

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1. The source contract and the common argument

This paper gives a common proof of the three principal results of Hou, Feng, Leemans and Qu, *String C-groups of order $4p^m$* [HFLQ]. The primary edition used here is arXiv:2407.10388v1, comprising twelve pages. Its Theorems 1.1, 1.2 and 1.3, with the construction in Theorem 5.1, concern nondegenerate regular polyhedra of order $4p^m$, where p is an odd prime. We recover the structural theorem, classify the abelian case, and realize every member of the nonabelian family. A displayed marking in Theorem 1.2 of that edition requires a correction, given in §4. We do not assert that the later journal edition contains the same error.

The proof has two parts. First, the odd Sylow subgroup is generated by one orbit under a Klein four group. If this subgroup is abelian, precisely two characters can occur; the intersection property then determines their orders. Second, a truncated cyclotomic module realizes the nonabelian construction. Complex conjugation supplies one reflection and additive negation supplies the other. This also identifies, generator by generator, the presentation used in [HFLQ]. The character decomposition is a reworking of [HFLQ, Lemma 2.4 and §4]. The cyclotomic organization and its direct presentation bridge are the replacement proof supplied here; no historical priority claim for this model is made.

We retain the odd/even part of Cunningham–Pellicer's classification of tight regular polyhedra [CP, Theorem 1.1] in one converse. That theorem is already an explicit input to [HFLQ]. Sylow's theorem, Lagrange's theorem, elementary finite abelian group theory, and Eisenstein's irreducibility criterion are standard background. The proof below does not require a classification of groups of maximal class or of Hughes subgroups.

A **marked string C-group of rank three** is a finite group generated by three distinct involutions r_0, r_1, r_2, with $r_0 r_2 = r_2 r_0$, such that

$$G_I \cap G_J = G_{I\cap J}, \qquad G_I = \langle r_i : i \in I\rangle, \quad I, J \subseteq \{0,1,2\}.$$

Its type is $\{k_1, k_2\}$, where $k_1 = |r_0 r_1|$ and $k_2 = |r_1 r_2|$. Nondegeneracy means $k_1, k_2 \geq 3$. Duality reverses the marking. We call the marking tight if $|G| = 2k_1k_2$. Isomorphisms of marked groups preserve the three generators. The standard correspondence with abstract regular polyhedra is retained when translating these statements into geometric language.

Theorem A (the odd kernel). Suppose that p is odd, $m \geq 2$, and $(G; r_0, r_1, r_2)$ is a nondegenerate string C-group of order $4p^m$. Its Sylow p-subgroup P is normal, and

$$G = P \rtimes K, \qquad K = \langle r_0, r_2 \rangle \cong C_2 \times C_2.$$

Both type parameters are divisible by p, at least one is divisible by $2p$, and the minimum number of generators of P is two.

Theorem B (the abelian kernel). Under the hypotheses of Theorem A, P is abelian if and only if the marking is tight. Up to duality, all such markings are the following, one for each $1 \le a \le b$ with $a + b = m$:

$$P = \langle x \rangle \times \langle y \rangle, \quad |x| = p^b, \quad |y| = p^a,$$
$$x^{r_0} = x, \quad y^{r_0} = y^{-1}, \quad x^{r_2} = x^{-1}, \quad y^{r_2} = y^{-1}, \quad r_1 = xyr_2.$$

The type is $\{2p^a, p^b\}$, and for this type the marked group is unique. Here and below $x^g = g^{-1}xg$.

Theorem C (the nonabelian family). For every odd prime p and every $m \ge 3$, there is a string C-group of order $4p^m$ and type $\{p, 2p\}$ whose Sylow p-subgroup is nonabelian. Put

$$O = \mathbb{Z}[\zeta], \quad \zeta = \zeta_p, \quad \pi = \zeta - 1, \quad A = O/\pi^{m-1}O.$$

Let $P = A \rtimes \langle \beta \rangle$, where $|\beta| = p$ and $v^\beta = \zeta v$. Define

$$\sigma(v) = \overline{v}, \quad \sigma(\beta) = \beta^{-1}, \qquad \tau(v) = -v, \quad \tau(\beta) = \beta.$$

Then $G = P \rtimes \langle \sigma, \tau \rangle$ with marking

$$r_0 = 1_A \tau\sigma, \qquad r_1 = \beta\tau\sigma, \qquad r_2 = \sigma$$

has the asserted properties. The symbol 1_A means the additive class of the ring element 1, not the identity of the additive group. This is exactly the marked presentation of [HFLQ, Theorem 5.1], under $s_i \mapsto \pi^{i-1}$.

2. Intersection and orbit lemmas

Lemma 2.1 (the rank-three test). Let three distinct involutions generate G, let r_0 commute with r_2, and suppose both adjacent products have order at least three. The full intersection property is equivalent to

$$\langle r_0, r_1 \rangle \cap \langle r_1, r_2 \rangle = \langle r_1 \rangle. \tag{2.1}$$

Proof. Necessity is immediate. For sufficiency, put $H_{ij} = \langle r_i, r_j \rangle$. The subgroup H_{02} consists of $1, r_0, r_2, r_0r_2$. If its intersection with H_{01} contains either r_2 or r_0r_2, then $r_2 \in H_{01} \cap H_{12} = \langle r_1 \rangle$, contrary to distinctness. Thus $H_{02} \cap H_{01} = \langle r_0 \rangle$; dually $H_{02} \cap H_{12} = \langle r_2 \rangle$. These three equalities also show that no generator belongs to the subgroup on the other two. Intersections involving singletons now follow, while those involving the empty or full index set are immediate. This covers every pair of subsets. $\square$

In particular, for a string C-group,

$$2k_1k_2 = |H_{01}H_{12}| \leq |G|. \tag{2.2}$$

The equality uses $|H_{01}| = 2k_1$, $|H_{12}| = 2k_2$ and (2.1).

Lemma 2.2 (prime-index intersection). Let $C \leq H \cap J$ be subgroups, with $[H : C]$ prime. If $H \not\leq J$, then $H \cap J = C$.

Proof. The tower formula gives $[H : C] = [H : H \cap J][H \cap J : C]$. If the first factor were one, then $H \leq J$. Therefore the second is one. □

Lemma 2.3 (one inverted orbit). Suppose $G = P \rtimes K$, where P has odd order and $K = \langle a, b\rangle$ is a Klein four group. Suppose $G = \langle K, r\rangle$ with r an involution. Write $r = ut$, with $u \in P$ and $t \in K$. Then $t \neq 1$, $u^t = u^{-1}$, and for either $k \in K \setminus \langle t\rangle$,

$$P = \langle u, u^k\rangle. \tag{2.3}$$

Proof. An odd-order group contains no involution, so $t \neq 1$. The equation $r^2 = 1$ gives $u^t = u^{-1}$. The subgroup $L = \langle u, u^k\rangle$ is stable under t, which inverts these two generators, and under k, which exchanges them. Consequently LK is a subgroup containing K and r. It is G, and intersection with P gives $L = P$. □

The last statement concerns the two specified generators; it does not assert that inversion is an automorphism of an arbitrary nonabelian P.

3. Proof of the structural theorem

Let n_p be the number of Sylow p-subgroups. Sylow's theorem gives $n_p \mid 4$ and $n_p \equiv 1 \pmod p$. The only possibility other than $n_p = 1$ is $p = 3$, $n_p = 4$. In that case conjugation on the four Sylow subgroups gives a homomorphism $G \longrightarrow S_4$. Its kernel N lies in $N_G(P) = P$, since the normalizer has index four. Thus the image has order $4 \cdot 3^j$ and divides 24, so $j = 0$ or 1. If $j = 0$, then $N = P$, making P normal, a contradiction. If $j = 1$, the image is the subgroup A_4 of S_4. But the involutions of A_4 all lie in its normal Klein four subgroup, so they cannot generate A_4. The image of an involution-generated group is generated by involutions or identities. This again is a contradiction. Hence P is normal.

The endpoint subgroup $K = \langle r_0, r_2\rangle$ has order four by the intersection property. Its intersection with P is trivial. Since $|PK| = 4p^m$, we have $G = P \rtimes K$.

Each adjacent dihedral subgroup has order $2k_i$, so $k_i \mid 2p^m$. Nondegeneracy therefore implies $p \mid k_i$. If both k_i were odd, their images in $G/P = K$ would force $r_0P = r_1P = r_2P$, contradicting the distinct images of the endpoints. At least one k_i is even. Lemma 2.3 shows that P has at most two generators. If P were cyclic, the nontrivial p-parts of both adjacent rotation subgroups would contain its unique subgroup of order p. Their intersection would then be nontrivial of odd order, contradicting (2.1). Thus P is not cyclic and needs exactly two generators. This proves Theorem A.

4. The two-character classification

Assume that P is abelian, and temporarily write its operation additively. For commuting involutions a, b on an abelian group of odd order, division by four is a well-defined automorphism. The four operators

$$e_{\epsilon,\delta} = \tfrac{1}{4}(1+\epsilon a)(1+\delta b), \qquad \epsilon, \delta \in \{1, -1\}, \tag{4.1}$$

are orthogonal idempotents whose sum is the identity. This follows by expanding the products and using $a^2 = b^2 = 1$, $ab = ba$. Thus P is the direct sum of their images, on which a, b act respectively by ϵ, δ.

Write $r_1 = ut$ as in Lemma 2.3. Since t inverts u and commutes with K, it inverts every generator in the K-orbit of u, and hence all of P. Only the two characters χ of K with $\chi(t) = -1$ can occur. For either of them, the projection of every orbit element is plus or minus $e_\chi u$. Therefore its component is cyclic, generated by this projection. Both components are nonzero: otherwise P would be cyclic. This argument justifies the two-component conclusion without deducing it merely from the number of generators of P.

Return to multiplicative notation. If $t = r_0 r_2$, the two components have characters $(+,-)$ and $(-,+)$. Choose their generators x, y so that $u = xy$. Direct multiplication gives adjacent orders $2|y|$ and $2|x|$. Formula (2.2) would imply

$$8|x||y| \le |G| = 4|x||y|,$$

which is impossible. Hence t is an endpoint generator. By duality take $t = r_2$. The two components now have characters $(+,-)$ and $(-,-)$, so their actions and $r_1 = xyr_2$ are precisely those in Theorem B. Writing $|x| = p^b$, $|y| = p^a$, we obtain

$$|r_0 r_1| = 2p^a, \qquad |r_1 r_2| = p^{\max(a,b)}.$$

Applying (2.2) once more gives $a + \max(a,b) \le a + b$, and hence $a \le b$. Equality holds in (2.2); the marking is tight.

Conversely, take the group and marking displayed in Theorem B. Both r_0 and r_2 invert y, while only r_2 inverts x, so the three marked elements are involutions and the endpoints commute. The elements $u = xy$ and $u^{r_0} = xy^{-1}$ generate x, y: their product and quotient give x^2, y^2, and squaring is invertible on either cyclic odd-order group. Thus the marking generates G.

The adjacent orders are $2p^a, p^b$. The intersection of H_{01} with P is $\langle y \rangle$: its image in K has order four and its rotation square generates $\langle y \rangle$. Similarly $H_{12} \cap P = \langle xy \rangle$, and its image in K is $\langle r_2 \rangle$. The groups $\langle y \rangle$ and $\langle xy \rangle$ meet trivially. Indeed, $x^j = 1$ forces $p^b \mid j$, and then $y^j = 1$ because $a \le b$. It follows that the homomorphism $H_{01} \cap H_{12} \to K$ is injective with image contained in $\langle r_2 \rangle$. The intersection contains r_1, so it is exactly $\langle r_1 \rangle$. Lemma 2.1 proves the full intersection property. The two cyclic component orders and their marked generators determine the marked group, proving uniqueness in the abelian case.

It remains to prove that a tight marking has an abelian kernel. By Theorem A its type parameters have the form p^b or $2p^b$. Tightness and $|G| = 4p^m$ force exactly one parameter to be even, and

the two exponents sum to m. The retained theorem [CP, Theorem 1.1(b),(c)] says that in this odd/even situation, excluding the parameter 4, tightness requires that the even parameter divide twice the odd one, and that the marked polyhedron is unique. The even parameter here is $2p^a$, never 4. Thus $a \le b$. The model already constructed is tight of that type, so uniqueness identifies every tight marking with it. This proves the converse and completes Theorem B.

The literal marking in the primary edition. The displayed presentation in [HFLQ, Theorem 1.2, physical p.2] has $x^{r_0} = x$, $y^{r_0} = y^{-1}$ but writes $r_1 = xyr_0$. Its square is x^2, not the identity, since $|x|$ is a positive power of the odd prime p. The body of [HFLQ, §4, physical p.6] uses xyr_2 in the corresponding character case. Our model uses that marking and assigns the larger exponent to x. The corrected statement, rather than the impossible literal formula, is what Theorem B recovers. No open conjecture is proved by this correction.

5. The cyclotomic lattice

Let p be an odd prime, $O = \mathbb{Z}[\zeta_p]$ and $\pi = \zeta_p - 1$. The polynomial

$$\Phi_p(1+X) = p + \binom{p}{2}X + \cdots + \binom{p}{p-1}X^{p-2} + X^{p-1}$$

is Eisenstein at p. In particular O is an integral domain with $\mathbb{Z}$-basis $1, \pi, \ldots, \pi^{p-2}$. Also $O/\pi O \cong \mathbb{F}_p$, by setting $\zeta = 1$ in Φ_p. Multiplication by π^j identifies $O/\pi O$ with $\pi^j O/\pi^{j+1}O$: injectivity follows by cancellation in the domain. It follows that

$$|O/\pi^n O| = p^n \qquad (n \ge 1). \tag{5.1}$$

For use in the precise presentation bridge, we record the entire additive lattice, not just its order. The identity

$$p = \prod_{j=1}^{p-1}(1-\zeta^j)$$

implies $pO = \pi^{p-1}O$. To justify equality of these ideals, each ratio $(1-\zeta^j)/(1-\zeta)$ lies in O and is a unit. Indeed, the automorphism $\zeta \mapsto \zeta^j$ shows that numerator and denominator both have norm p; the ratio has norm one. Multiplication by it is an integer matrix of determinant one, so its inverse is an integer matrix and the ratio is a unit.

For $1 \le r \le p-1$, the lattice $\pi^r O$ is

$$p\mathbb{Z}\,1 + \cdots + p\mathbb{Z}\,\pi^{r-1} + \mathbb{Z}\,\pi^r + \cdots + \mathbb{Z}\,\pi^{p-2}. \tag{5.2}$$

Every displayed generator belongs to $\pi^r O$; the displayed lattice has index p^r, equal to (5.1), so inclusion is equality. Write uniquely

$$n = (e-1)(p-1) + r, \qquad e \ge 1, \quad 1 \le r \le p-1.$$

Since $\pi^n O = p^{e-1}\pi^r O$, (5.2) gives the additive decomposition

$$O/\pi^n O \cong (\mathbb{Z}/p^e\mathbb{Z})^r \oplus (\mathbb{Z}/p^{e-1}\mathbb{Z})^{p-1-r}, \tag{5.3}$$

with the indicated basis classes $1, \pi, \dots, \pi^{p-2}$. A factor with modulus one is interpreted as trivial. This supplies both an exact finite coordinate interface and the individual orders of every s_i in the source presentation.

Conjugation preserves the ideal $\pi^n O$, since $\overline{\pi} = -\zeta^{-1}\pi$. Multiplication by ζ has order p on $O/\pi^n O$ when $n \geq 2$: it is not the identity, because $\pi \notin \pi^n O$, while its p-th power is the identity. Finally,

$$1 + \zeta + \cdots + \zeta^{p-1} = 0. \tag{5.4}$$

For each j not divisible by p, replacing ζ by ζ^j gives the same zero sum.

6. Proof of the nonabelian construction

Put $n = m - 1 \geq 2$ and $A = O/\pi^n O$. To specify the conjugation convention without ambiguity, realize P as pairs $(v, k) \in A \times \mathbb{Z}/p\mathbb{Z}$ with multiplication

$$(v, k)(w, l) = (v + \zeta^{-k} w, k + l). \tag{6.1}$$

Then $\beta = (0, 1)$ satisfies $v^\beta = \zeta v$. Equation (5.1) gives $|P| = p^m$. It is nonabelian because multiplication by ζ is nontrivial. Every element outside A has order exactly p: for $k \neq 0$, its p-th power has first coordinate $\sum_{i=0}^{p-1} \zeta^{-ik} v = 0$ by (5.4), and its image in P/A has order p.

The maps

$$\sigma(v, k) = (\overline{v}, -k), \qquad \tau(v, k) = (-v, k)$$

preserve (6.1), because conjugation sends ζ^{-k} to ζ^k. They are commuting involutory automorphisms. They are distinct and independent: σ inverts the nontrivial element β, whereas τ fixes it; τ is nontrivial on the nonzero odd-order group A. Thus they generate a Klein four group and $|G| = 4p^m$.

Set $s = 1_A$ and $h = \tau\sigma$. We have $s^h = s^{-1}$ and $\beta^h = \beta^{-1}$, in multiplicative notation inside P. Hence $r_0 = sh$, $r_1 = \beta h$ and $r_2 = \sigma$ are involutions. Also

$$r_0 r_2 = s\tau = r_2 r_0, \qquad r_0 r_1 = s\beta^{-1}, \qquad r_1 r_2 = \beta\tau.$$

The second product is outside A and has order p. The last has order $2p$, because β and τ commute and have coprime orders $p, 2$. These computations also show that the three involutions are distinct.

They generate G. Indeed, $r_1 r_2 = \beta\tau$ has p-th power τ; it then yields β. Together with r_2 this gives σ, and r_0 gives s. The β-conjugates of s are the classes of $1, \zeta, \dots, \zeta^{p-1}$, which additively generate A.

Finally let $H = \langle r_0, r_1 \rangle$ and $J = \langle r_1, r_2 \rangle$. Their orders are $2p$ and $4p$. The subgroup $C = \langle r_1 \rangle$ has prime index p in H. If $H \leq J$, then all three generators lie in J, forcing $G = J$, contrary to $4p^m > 4p$. Lemma 2.2 gives $H \cap J = C$, and Lemma 2.1 proves the intersection property. This proves Theorem C.

7. The exact source-presentation bridge

Use the integers e, r of (5.3) with $n = m - 1$, and let $s_i = \pi^{i-1}$ for $1 \leq i \leq p-1$. Their orders are p^e for $i \leq r$ and p^{e-1} for $i > r$. All s_i commute. Our convention $[x, y] = x^{-1}y^{-1}xy$ gives

$$[s_i, \beta] = \pi s_i = s_{i+1} \quad (1 \leq i < p-1).$$

The cyclotomic equation $\Phi_p(1 + \pi) = 0$ gives the final relation

$$[s_{p-1}, \beta] = \prod_{i=1}^{p-1} s_i^{-\binom{p}{i}}. \tag{7.1}$$

Thus every relation in [HFLQ, §5, physical p.6] holds in our group. Conversely, in the abstract group defined by that presentation, the subgroup on the commuting s_i has order at most p^{m-1}. The conjugation relations make it invariant under conjugation by β: inclusion is equality because this subgroup is finite and conjugation is injective. The quotient is generated by β of order dividing p. Therefore the presented group has order at most p^m. The surjection to our group of order p^m is consequently an isomorphism. It sends every named generator to the stated ring class, not merely to some group with the same order.

Under this isomorphism, σ fixes s_1 and inverts β, and τ inverts s_1 and fixes β. These are exactly the maps and markings of [HFLQ, Theorem 5.1]. More generally,

$$s_i^{\sigma} = (-1)^{i-1}\zeta^{1-i}\pi^{i-1} = \begin{cases} s_i^{\beta^{1-i}} & i \text{ odd}, \\ (s_i^{\beta^{1-i}})^{-1} & i \text{ even}. \end{cases} \tag{7.2}$$

In (7.2) the middle expression is additive ring notation and the right side is group notation. It follows from $\overline{\pi} = -\zeta^{-1}\pi$ and also holds for the additional commutators s_p, s_{p+1} used by the source. Expanding $\zeta^{p+1-i} = (1 + \pi)^{p+1-i}$ gives the binomial formula of [HFLQ, Lemma 5.2(1)]. Formula (7.2) gives Lemma 5.2(2), while the automorphism proof in §6 gives the content of Lemma 5.3. Thus the presentation and its involutory automorphisms are recovered directly, including parameters for which some listed s_i are trivial.

For completeness, $[A, \beta] = \pi A$, and induction gives $\gamma_j(P) = \pi^{j-1}A$ for $j \geq 2$. These groups are nontrivial through $j = n$ and trivial at $j = n + 1$. Hence P has nilpotency class $n = m - 1$, as in the source's maximal-class description. The motivating condition $A = H_p(P)$ on a Hughes subgroup is not needed: when $n \leq p - 1$, (5.3) makes A elementary abelian, all elements of P have order dividing p, and $H_p(P) = 1$. This explains why that condition should not be imposed on every parameter of the construction itself.

8. Attribution, verification, and limits

Theorem A is [HFLQ, Theorem 1.1], with its Sylow argument and inverted orbit made explicit. The abelian classification is [HFLQ, Theorem 1.2] with the literal marking corrected; the two-character proof retains the classical idempotent decomposition underlying its Lemma 2.4. The tightness converse retains precisely [CP, Theorem 1.1(b),(c)] and the exclusion of its exceptional parameter 4. Theorem C and the marked presentation are [HFLQ, Theorems 1.3 and 5.1]. Sections 5–7 replace its binomial verification and presentation-existence input by one cyclotomic lattice argument. These attributions concern the mathematical results; the replacement proof does not claim their discovery.

The accompanying finite checker uses the mixed-modulus coordinates (5.3), the companion matrix of $\Phi_p(1+X)$, and literal permutation actions. It verifies the displayed matrices, source relators, generator orders and all 64 subset-intersection equalities for each selected finite model. Its checks are examples of the uniform written proof, not a substitute for the arbitrary-p, m argument. It also checks the abelian marking and rejects the literal faulty formula.

The accompanying Lean file verifies the prime-index intersection argument and elementary reflection-product identities used in §6. Cyclotomic arithmetic, Sylow normality, the character classification, and the retained tight-polyhedron classification remain written proof or retained input; they are not represented as established Lean facts. The replay report states the exact tested parameters and formal theorem names. All required mathematical programs and dependencies must be charged in any later comparison. No compression ratio is asserted here.

The complete principal contract is the three main theorems, with their explicit marked construction and the stated correction. We do not reproduce the introductory numerical census of small soluble groups, the historical survey, or the general classification of Hughes groups. We solve no open conjecture in this application. Independent review is required before any principal-source coverage is admitted.

References

[HFLQ] D.-D. Hou, Y.-Q. Feng, D. Leemans and H.-P. Qu, *String C-groups of order* $4p^m$, arXiv:2407.10388v1 (15 July 2024), 12 pp.; later published in *Communications in Algebra* 53 (2025), 3388–3399. The bound local primary is `sources/regular_4pm2025.pdf`.

[CP] G. Cunningham and D. Pellicer, *Classification of tight regular polyhedra*, arXiv:1408.6703v2, Theorem 1.1, physical p.2; *Journal of Algebraic Combinatorics* 43 (2016), 665–691. Primary theorem accessed at https://arxiv.org/pdf/1408.6703v2. Only the explicitly stated theorem is retained; no whole-source rederivation or locally bound PDF is claimed.